Sai Charan Ajjarapu

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.60
FIITJEE Junior College	Class XII (Telangana State Board)	2023	98.30%
FIITJEE World School	Class X (Telangana State Board)	2021	10.00%

SCHOLASTIC ACHIEVEMENTS

AIR 39 in **ISI**(Indian Statistical Institute) Entrance Test. Was offered a seat in the B. Stat Honors program.

Secured **AIR 777(Top 0.04%)** in the Kishore Vaigyanik Protsahan Yojana(KVPY, SA Stream) held by Govt. of India.

AIR 380 in JEE Mains | **AIR 1030** in JEE Advanced

Cleared **National Talent Search Examination(NTSE)**, securing a **scholarship** offered by **Government of India**

WORK EXPERIENCE

Quantitative Research Intern, 26 Miles Capital

- Built walk-forward pipelines for 3 strategies to predict optimal daily options strategy for NIFTY using only first-30-min data; achieved 64% directional accuracy vs. 33.3% random baseline across 4+ years of out-of-sample trading days.
- Explored regime classification via PCA, K-means and lagged relation across 10+ features and pivoted to a feature-quintile-based next-day prediction approach, improving pre-existing strategy by 5% (PnL), 18% (MDD), and Sharpe.
- Decomposed options PnL into Greek contributions and GEX percentiles, and designed a GEX-adaptive allocation scheme that improved risk-adjusted PnL over equal-weighting.
- Worked on delta-hedging in a no-transaction-band, reducing expected rebalancing costs while preserving band width.

Research Intern, Dept. of Mathematics, IIT Hyderabad

Working on Stochastic Differential Equations, and applications for Finance.

PROJECTS

Option Pricing with Fast Fourier Transform

- Implemented Carr-Madan FFT algorithm to price European options under Black-Scholes and Variance gamma models.
- Cross validated FFT prices against analytical and direct integration benchmarks(± 0.0007 accuracy), achieving a 42.9X speedup by reducing complexity to O(N log N).

Agentic Linux Diagnostic Assistant

- Built an LLM agent that diagnoses Linux/Service/Git issues via 18 types tools exposed through a custom MCP server, orchestrated by Gemini & LangGraph.
- Designed a risk-tiered permission system with human-in-the-loop approval to safely constrain LLM driven system actions.

Generalised Hyperplane Trees | *Data Structures and Applications Lab project*

- Developed a Hyperplane Tree structure in C++ from scratch with multiple variants and used it to cluster real world data, resulting in 93% accuracy. Analyzed classification, build, and test accuracies and other parameters.

Product ID Tracker using BSTs and Hash Tables | *Data Structures and Algorithms coursework*

- Implemented end-to-end tracking system for inventory validation using custom built hash tables and BSTs in C.
- Simulated real-world defect identification using search operations and tree balancing.

Statistical and Data Analysis | *Applied Statistics Project*

- Performed EDA on multiple real world datasets, and applied hypothesis testing, confidence intervals, chi-squared tests, MLE, MOM estimators to extract inferences.
- Applied Central Limit Theorem to data obtained from Dominos outlet, and derived estimations for average orders and sales.

SKILLS

Programming Languages : C, C++, Python, R, SQL

Tools : Git, Github, Jupyter Notebook, Latex, Linux

AI : GenAI, RAG, LLM, MCP, LangChain, LangGraph

Soft Skills : Team Work, Leadership, Communication, and Negotiation.

RELEVANT COURSES

Computer Science : Introduction to Programming, Data Structures, Algorithms, Compilers-I, DBMS-I.

Applied Mathematics : Linear Algebra, Probability, Applied Statistics, Convex Optimization, Fuzzy Logic Connectives.

Pure Mathematics : Calculus, Combinatorics & Graph Theory, Number Theory, Differential Equations, Transform Techniques, Groups & Rings, Analysis.

POSITIONS OF RESPONSIBILITY

Sponsorship Head, Elan & nVision (IIT Hyderabad's annual techno-cultural fest) **:** Was directly involved in raising sponsorship and managing a budget of upwards of eight figures, making the fest the biggest in IIT Hyderabad in terms of scale, footfall, budget.

Literary Cup Captain, IIT Hyderabad, Inter IIT Cultural Meet 7.0 : Led a team of 10 for all literary competitions at the 7th Inter IIT Cultural meet, securing IIT Hyderabad's best results in Literary Arts.

Active member of **Finance & Consulting Club**, **Blockchain Society**, and **Literary Society** of IIT Hyderabad.

EXTRACURRICULAR

Qualified several **math and science olympiads** including **PRMO, SOF**.

Ranked **26th** of 10,000+ teams from India and was invited to **IIT Guwahati** for **Technothlon** 2019(**math and reasoning contest**)

Certified in **Trinity Music College, London's Rock and Pop Guitar** program.